Chapter 3: Conditional Probability and Independence (Sections 3.4 – 3.5)

3.4 Independent Events

Definition

Two events E and F are **independent** if the occurrence of one does not affect the probability of the other.

- Mathematically: $P(E|F) = P(E)$
- Equivalent Condition (Symmetric):

$$P(EF) = P(E)P(F)$$

- If this condition does not hold, the events are **dependent**.

Examples

1. **Card Selection:**
   - E: Ace $(4/52)$. F: Spade $(13/52)$.
   - $P(EF)$ (Ace of Spades) $= 1/52$.
   - $P(E)P(F) = (4/52)(13/52) = 1/52$.
   - **Conclusion:** Independent.
2. **Dice Rolls:**
   - E: Sum is 6. F: First die is 4.
   - $P(E|F) = P(\text{Sum } 6 \mid \text{1st is } 4) = P(\text{2nd is } 2) = 1/6$.
   - $P(E) = 5/36$.
   - **Conclusion:** Dependent ($1/6 \neq 5/36$).
3. **Dice Sum 7:**
   - E: Sum is 7. F: First die is 4.
   - $P(E|F) = 1/6$. $P(E) = 6/36 = 1/6$.
   - **Conclusion:** Independent.

Propositions

- **Independence of Complements:** If E and F are independent, then E and F^c are also independent.
  - *Proof:* $P(EF^c) = P(E) - P(EF) = P(E) - P(E)P(F) = P(E)(1 - P(F)) = P(E)P(F^c)$.

Multiple Events

Three events E, F, G are independent if **all** conditions hold:

1. $P(EF) = P(E)P(F)$
2. $P(EG) = P(E)P(G)$
3. $P(FG) = P(F)P(G)$
4. $P(EFG) = P(E)P(F)P(G)$

- *Note:* Pairwise independence does not imply mutual independence (see Example 4e: Sum of dice is 7, First die is 4, Second die is 3).

Independent Trials

A sequence of subexperiments is independent if the outcome of any group does not affect the others.

- **Bernoulli Trials:** Success prob p, Failure $1-p$.
  - Prob of exactly k successes in n trials: $\binom{n}{k}p^k(1-p)^{n-k}$.
  - Prob of at least 1 success: $1-(1-p)^n$.

System Reliability

- **Parallel System:** Functions if at least one component works.
  - $P(\text{System works}) = 1 - \prod(1-p_i)$.
- **Series System:** Functions only if all components work.
  - $P(\text{System works}) = \prod p_i$.

Gambler's Ruin Problem

- **Scenario:** Gambler A starts with i, B with $N-i$. A wins 1 unit with prob p, loses 1 with prob $q=1-p$. Game ends when A hits N or 0.
- **Probability A wins (P_i):**
  - If $p=1/2$: $P_i = i/N$.
  - If $p \neq 1/2$: $P_i = \frac{1-(q/p)^i}{1-(q/p)^N}$.
- **Infinite Game:** If $p=1/2$, probability game goes on forever is 0.

3.5 P(·|F) is a Probability

Core Concept

Conditional probability $Q(E) = P(E|F)$ satisfies all three axioms of probability.

1. $0 \leq P(E|F) \leq 1$.
2. $P(S|F) = 1$.
3. If E_i are mutually exclusive, $P(\cup E_i|F) = \sum P(E_i|F)$.

Implication: Any theorem true for unconditional probability is true for conditional probability.

- $P(A \cup B|F) = P(A|F) + P(B|F) - P(AB|F)$.

Conditional Independence

Events E_1 and E_2 are conditionally independent given F if:

$$P(E_1E_2|F) = P(E_1|F)P(E_2|F)$$

- *Example:* Sequential updating of probabilities (Laplace's Rule, Coin Flips) relies on conditional independence of future trials given the true bias/parameter.

Laplace's Rule of Succession

- **Scenario:** $k+1$ coins, coin i has prob i/k of heads. Choose one at random.
- **Result:** If first n flips are heads, the prob the $(n+1)^{st}$ flip is heads is $\approx \frac{n+1}{n+2}$.

Matching Problem (Conditional Approach)

Using conditioning to solve for P_n (prob of no matches):

- Condition on whether first man picks his own hat (M) or not (M^c).
- Derives recurrence: $P_n = P_{n-1} - \frac{1}{n}(P_{n-1} - P_{n-2})$.
- Solution: $P_n = \sum_{i=0}^{n} \frac{(-1)^i}{i!}$.

1